

Report Information	
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Demo 1: Retargeting & Character Diversity

Screenshot(s) of feature:



Describe your implementation:

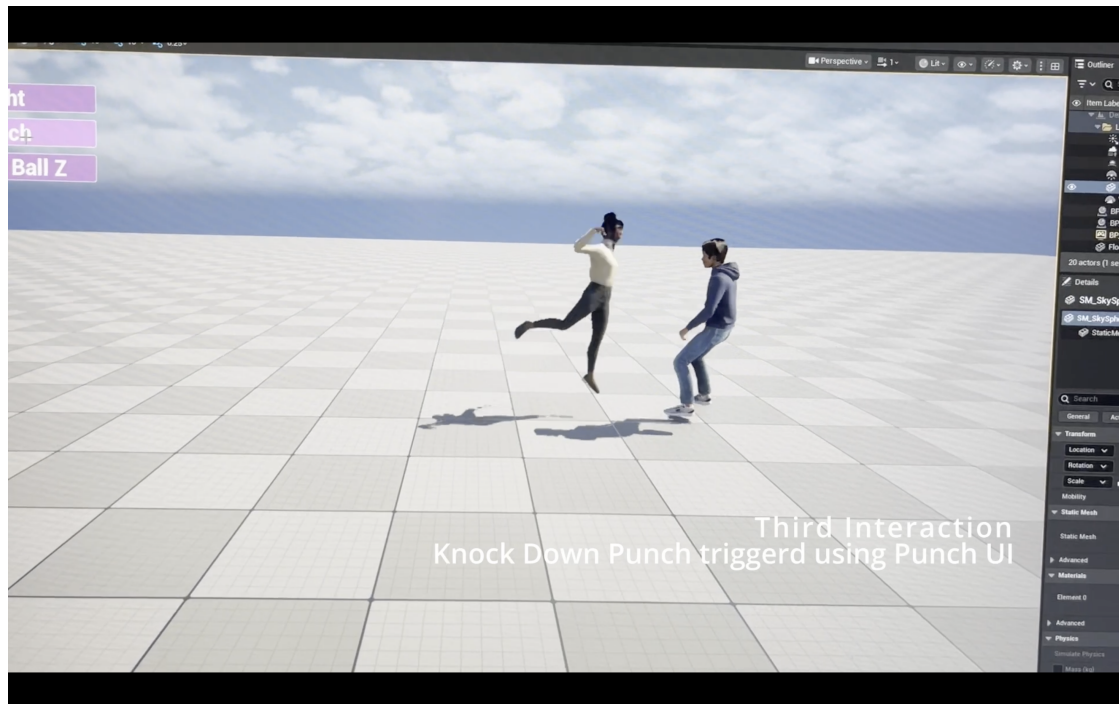
The three MetaHuman characters were imported into Unreal Engine using Quixel Bridge. To apply the provided motion capture data, an IK Retargeter was created for each MetaHuman, with the ViconMesh skeleton set as the source skeleton and the MetaHuman skeleton set as the target. The bone chains were then manually aligned to match the source skeleton's pose, ensuring correct correspondence between the two rigs. Once the retargeting was configured, the animation was applied to each MetaHuman by selecting the character in the viewport, navigating to the Body component in the Details panel, changing the Animation Mode to Animation Asset, and assigning the newly retargeted walking animation.

Critical evaluation:

The primary challenge encountered during this phase was the bone alignment process within the IK Retargeter. While Unreal Engine provides an Auto Align feature to automate this step, it proved insufficient in practice, several bones were joint rotations and misaligned limbs. These issues required manual correction on a per-bone basis, which was time consuming but necessary to achieve a clean result. This highlights a limitation of the auto-alignment tool when working with skeletons that differ significantly in proportions, such as the ViconMesh capture rig versus the MetaHuman skeleton.

Demo 2: Two-character Interaction

Screenshot(s) of feature:



Describe your implementation:

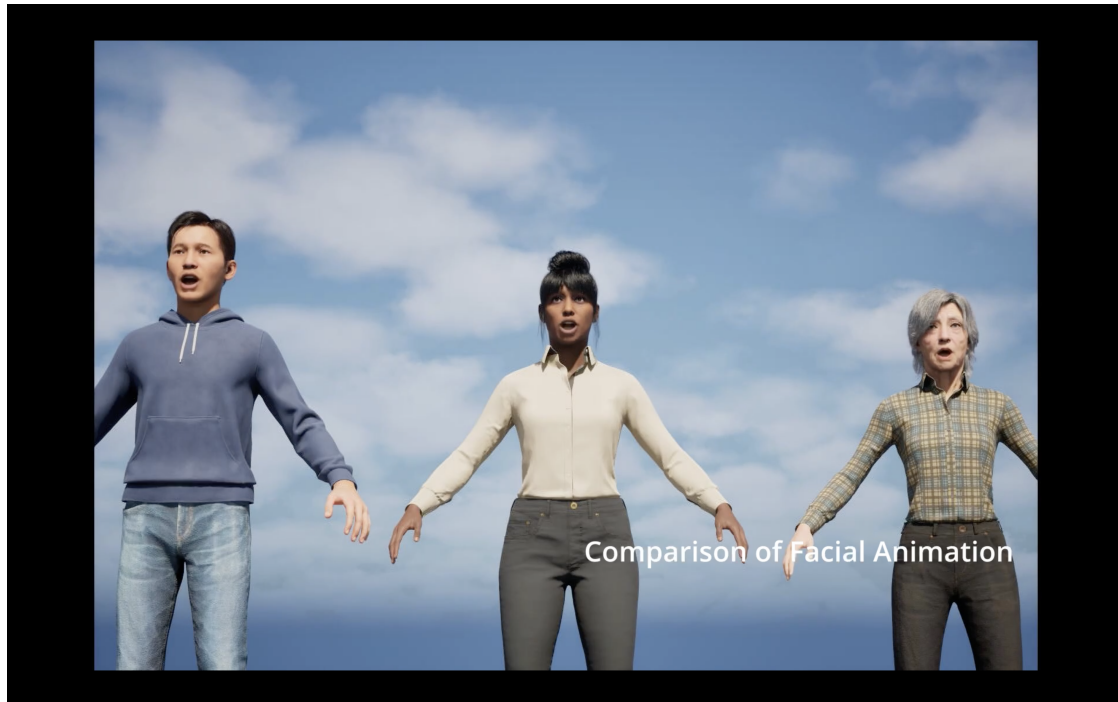
For the two-character interaction, three distinct animation sequences were implemented: a fight sequence, a knockout punch interaction, and a Dragon Ball Z pose recreation. The user interface was built using a Widget Blueprint in Unreal Engine, where a Canvas Panel was used to layout three buttons labeled Fight, Punch, and Dragon Ball Z. Each button was configured with an OnClicked() event, which upon being pressed, triggered the corresponding animation Blueprint logic. Within each Blueprint, the relevant animation was fired on the appropriate MetaHuman character, allowing the user to control and switch between the three interactions in real time.

Critical Evaluation:

The implementation of the two-character interaction was relatively straightforward, as Unreal Engine's Widget Blueprint system provides an intuitive workflow for connecting UI events to animation logic. The OnClicked() event binding made it simple to map each button directly to its corresponding animation trigger without requiring complex state management. The three interactions: a fight sequence, a knockout punch, and a Dragon Ball Z pose, provided a varied and visually engaging demonstration of user-triggered animation transitions.

Demo 3: Facial Animation

Screenshot(s) of feature:



Describe your implementation:

Facial animation was implemented using MetaHuman Animator driven by emotional video clips sourced from the RAVDESS dataset. The video clips were imported directly into Unreal Engine and processed using Live Link, which facilitated the transfer of facial performance data onto the MetaHuman face meshes. Each clip was targeted to the face mesh of the respective MetaHuman character, allowing the emotional performance captured in the source video to drive the facial bones and blend shapes of the digital avatar. The facial animation was applied to three MetaHuman characters: a young female, a male, and an aged female enabling a direct cross-character comparison of emotion transfer. To create the concatenated sequence, a Level Sequence was created in Unreal Engine's Sequencer, and the individual emotional animation clips neutral, happy, upset, and agitated were added sequentially to the actor's face track. The Face Control Rig track was removed from the Sequencer to prevent it from overriding the baked facial animation, which allowed the imported MetaHuman Animator clips to play back correctly in the correct order.

Critical Evaluation:

The MetaHuman Animator pipeline produced convincing results for broad, high-intensity emotions such as happiness and agitation, where the facial movements in the source video were pronounced and clearly readable. However, subtle emotions such as neutral and mild upset were transferred with less accuracy, as the system struggled to capture fine micro-expressions that are not strongly represented in the blend shape targets of the MetaHuman rig. Noticeable artefacts included occasional jaw instability and eye region jitter during transitions between emotional clips in the concatenated sequence, likely caused by the absence of smoothing between clip boundaries. A further limitation was the variation in transfer quality across the three MetaHuman characters while the young female avatar produced the most accurate results, the aged female MetaHuman exhibited more pronounced skin deformation artefacts due to differences in the rig's blend shape configuration.

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